BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) — Group - I — H.S.S.C (Part-II) A/2024 (For All Sessions)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8461

Note: Note: Write answers to the questions on the objective answer sheet provided. Four possible answers are given. Which answer you consider correct fill the corresponding circle A,B,C or D in front of each question with marker or ink on the answer sheet provided.

Q1. The excretory product that requires minimum water for its elimination as compared to others is:

(A) Urea
(B) Ammonia
(C) Uric acid
(D) Creatinin [sic]

Q2. Which of the following is bone of axial skeleton:

(A) Ribs
(B) Shoulder girdle
(C) Pelvis
(D) Femur

Q3. Cardiac muscles are :

(A) Voluntary
(B) Involuntary
(C) Both (A) and (B)
(D) None of these

Q4. Which one is not related to others is: [sic]

(A) Cretinism
(B) Myxoedema
(C) Exophthalmic goiter
(D) Diabetes mellitus

Q5. Gastrin is the hormone produced by:

(A) Gut
(B) Liver
(C) Pancreas
(D) Oral cavity

Q6. Reproduction is very important for the survival of:

(A) Species
(B) Population
(C) Individual
(D) Both (A) and (B)

Q7. For maximum growth of plants, the optimum temperature is:

(A) 15 - 20 °C
(B) 20 - 25 °C
(C) 25 - 30 °C
(D) 30 - 35 °C

Q8. Enzyme are responsible for assembly of: [sic]

(A) Nucleic acid
(B) Protein
(C) Carbohydrate
(D) All (A),(B) and (C)

Q9. In Bacteria, the newly synthesized mRNA is released in:

(A) Cytoplasm
(B) Nucleus
(C) Mitochondria
(D) Chloroplast

Q10. In Klinefelter's syndrome:

(A) One x. chromosome is missing
(B) Additional sex-chromosome is present
(C) One autosome is missing
(D) None of these

Q11. When a haemophilic carrier women marries a normal man, who among her offspring may be affected: [sic]

(A) All her children
(B) All her daughters
(C) Half of her daughters
(D) Half of her sons

Q12. A team of Japanese scientists is attempting to introduce the C_4 photosynthetic cycle into:

(A) Rice
(B) Wheat
(C) Corn
(D) Oat

Q13. It makes bacterial cell more permeable to take up recombinant plasmid:

(A) Sodium chloride
(B) Potassium chloride
(C) Calcium chloride
(D) Cesium chloride

Q14. Who published an essay on "The principle of population" ?

(A) Darwin
(B) Lyell
(C) Malthus
(D) Mendel

Q15. Bacteria and Fungi are examples of:

(A) Decomposer (B) Producer

(C) Consumer (D) Grazer

Q16. The light in which zone is insufficient to support photosynthesis:

(A) Littoral (B) Limnetic

(C) Profundal (D) All of these

Q17. Total area of world under cultivation is:

(A) 9 % (B) 10 %

(C) 11 % (D) 12 %

Biology (Subjective) — Group - I — H.S.S.C. (Part-II) A / 2024 (For All Sessions)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q2. Write short answers of any eight parts of the question. [2x8=16]

(i) Why color of plant leaves turns yellow in autumn?

(ii) How plants protect their enzyme from denaturation at high temperature?

(iii) Compare hydrophytes with xerophytes.

(iv) Out of 12 pairs of ribs, why only two pairs of ribs are called free floating ribs?

(v) Describe internal structure of cilium.

(vi) How low Ca+2 in blood affects bones in growing children?

(vii) Differentiate between chemotactic and chemotropic movements.

(viii) Name the cells found outside seminiferous tubules. Give one main function of those cells.

(ix) Why is there no productivity in profundal zone in aquatic ecosystem?

(x) What is Tundra? Does this ecosystem exist in Pakistan?

(xi) How combusion of fossil fuels is related to stone cancer? [sic]

(xii) Write down the two impacts of ozone layer depletion on human life.

Q3. Write short answers of any eight parts of the question. [2x8=16]

(i) How do plants respond to various stimuli under stress?

(ii) Define Threshold frequency to initiate nerve impulse.

(iii) What do you know about commercial applications of Gibberellins? (at least two).

(iv) Why is blood group "O" considered universal donor?

(v) What do you know about XX - XY mechanism of sex determination?

(vi) Define product rule. Give an example.

(vii) How cancer patients are being treated by gene therapy?

(viii) Give two practical uses of DNA finger printing technology.

(ix) What are restriction endonucleases? Give example.

(x) How does length of food chain affect an ecosystem?

(xi) What is pyramid of energy?

(xii) Define Autecology. Give example.

Q4. Write short answers of any six parts of the question. [2x6=12]

(i) The plant cell size increase in number of cells and flowering are affected by light. How?

(ii) Differentiate between Gastrula and Neurula.

(iii) Compare the homologous and analogous organs.

(iv) How a particular amino acid is brought at a specific ribosomal site? Give the role of enzyme also.

(v) What is point mutation? Write one example.

(vi) Draw the structural formulae of Adenine and Guanine.

(vii) Why Anaphase is considered critical phase?

(viii) How cancer cells are different from normal cells?

(ix) Write any two points of Lamarckism.

Q. Section - II

Note: Attempt any three questions from the following: (8x3=24)

Q5. (a) What is Renal failure? Describe its treatment. [4]

(b) How does cytokinesis occur in animal cells? In which way does it differ from that in plant cell? [4]

Q6. (a) Discuss genetic and hormonal causes about deformities of skeleton. [2+2]

(b) Explain Nitrogen cycle with the help of its sketch? [4]

Q7. (a) Which factors are involved in establishment of resting membrane potential? Explain. [4]

(b) Define Hardy-Weinberg Theorem. How its equation in used to calculate allele & genotype frequency? [sic] [4]

Q8. (a) Discuss sex determining pattern in grass hopper and birds. [4]

(b) Describe female reproductive cycle in human. [4]

Q9. (a) What is growth? Discuss its phases in plants. [4]

(b) Write a note on transgenic animals. [4]

GROUP – II

Biology (Objective) — Group - II — H.S.S.C (Part-II) A/2024 (For All Sessions)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8462

Note: Note: Write answers to the questions on the objective answer sheet provided. Four possible answers are given. Which answer you consider correct fill the corresponding circle A,B,C or D in front of each question with marker or ink on the answer sheet provided.

Q1. Which part of brain acts as homeostatic thermostat in human?

(A) Thalamus
(B) Hypothalamus
(C) Cerebrum
(D) Medulla

Q2. Most of cartilage consists of:

(A) Osteoclasts
(B) Osteocytes
(C) Chondrocytes
(D) Cartilocytes [sic]

Q3. Commercial cork is obtained from wood of:

(A) Quercus suber
(B) Dalbergia sisso
(C) Solanum nigrum
(D) Cassia fistula

Q4. Which hormone promotes flowering in pineapple?

(A) Auxins
(B) Cytokinins
(C) Ethene
(D) Abscisic acid

Q5. In an unstimulated neuron, the membrane potential is approximately:

(A) + 50 mV
(B) - 50 mV
(C) + 70 mV
(D) - 70 mV

Q6. Preparation for Lactation is stimulated by:

(A) FSH
(B) ICSH
(C) LTH
(D) TSH

Q7. The cavity formed between somatic and splanchnic mesoderm is called:

(A) Archenteron
(B) Coelom
(C) Neurocoel
(D) Blastocoel

Q8. In eukaryotic cells, RNA polymerase - II makes:

(A) m-RNA
(B) r-RNA
(C) t-RNA
(D) c-DNA

Q9. A typical chromosome may contain ---------- nucleotides.

(A) 4 Billion
(B) 140 Billion
(C) 100 Million
(D) 140 Million

Q10. During cell cycle, chromosomal contents are doubled in:

(A) G_1. phase
(B) G_0 phase
(C) S-phase
(D) G_2-phase

Q11. Secreter gene "SE" is located on chromosome No: [sic]

(A) 11
(B) 19
(C) 21
(D) 23

Q12. Bacteria naturally contain restriction endonucleases for their protection against:

(A) Antibiotics
(B) Heavy metals
(C) Viruses
(D) Other bacteria

Q13. In Sanger's method dideoxyribonucleoside triphosphate is used to terminate the synthesis of:

(A) RNA
(B) DNA
(C) Protein
(D) Lipids

Q14. Which scientist proposed endosymbiont hypothesis?

(A) Margulis
(B) Cuvier
(C) Darwin
(D) Malthus

Q15. Bacteria present in root nodules fix nitrogen and convert into:

(A) Nitrites
(B) Nitrates
(C) Amino acids
(D) Ammonia

Q16. The productivity in tropical grasslands is more than :

(A) 1500 g/m²
(B) 2000 g/m²
(C) 3000 g/m²
(D) 4000 g/m²

Q17. Establishment of forests where no forest existed previously is called:

(A) Deforestation
(B) Afforestation
(C) Reforestation
(D) Forestation